## 6.8.4 Equation of a plane passing through three given non-collinear points

### (a) Parametric form of vector equation

**Theorem 6.17**

If three non-collinear points with position vectors $\vec{a},\vec{b},\vec{c}$ are given, then the vector equation of the plane passing through the given points in parametric form is

$$\vec{r}=\vec{a}+s(\vec{b}-\vec{a})+t(\vec{c}-\vec{a}),\ \text{where}\ \vec{b}\neq\vec{0},\ \vec{c}\neq\vec{0}\ \text{and}\ s,\ t\in\mathbb{R}.$$

**Proof**

Consider a plane passing through three non-collinear points $A,B,C$ with position vectors $\vec{a},\vec{b},\vec{c}$ respectively. Then atleast two of them are non-zero vectors. Let us take $\vec{b}\neq\vec{0}$ and $\vec{c}\neq\vec{0}$. Let $\vec{r}$ be the position vector of an arbitrary point $P$ on the plane. Take a point $D$ on $AB$ (produced) such that $\overrightarrow{AD}$ is parallel to $\overrightarrow{AB}$ and $\overrightarrow{DP}$ is parallel to $\overrightarrow{AC}$. Therefore,

$$\overrightarrow{AD}\ =\ s(\vec{b}-\vec{a}),\ \overrightarrow{DP}=t(\vec{c}-\vec{a}).$$

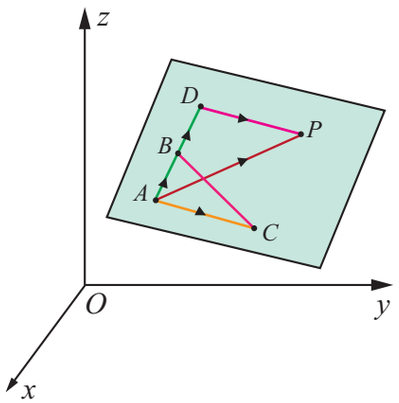



Fig. 6.27

Now, in triangle $ADP$, we have

$$\overrightarrow{AP}\ =\ \overrightarrow{AD}+\overrightarrow{DP}\ \text{or}\ \vec{r}-\vec{a}=s(\vec{b}-\vec{a})+t(\vec{c}-\vec{a}),\ \text{where}\ \vec{b}\neq\vec{0},\ \vec{c}\neq\vec{0}\ \text{and}\ s,\ t\in\mathbb{R}$$

. That is, $\vec{r}\ =\ \vec{a}+s(\vec{b}-\vec{a})+t(\vec{c}-\vec{a})$.

This is the parametric form of vector equation of the plane passing through the given three non-collinear points. ■

### (b) Non-parametric form of vector equation

Let $A,B,$ and $C$ be the three non collinear points on the plane with position vectors $\vec{a},\vec{b},\vec{c}$ respectively. Then atleast two of them are non-zero vectors. Let us take $\vec{b}\neq\vec{0}$ and $\vec{c}\neq\vec{0}$. Now $\overrightarrow{AB}=\vec{b}-\vec{a}$ and $\overrightarrow{AC}=\vec{c}-\vec{a}$. The vectors $(\vec{b}-\vec{a})$ and $(\vec{c}-\vec{a})$ lie on the plane. Since $\vec{a},\vec{b},\vec{c}$ are non-collinear, $\overrightarrow{AB}$ is not parallel to $\overrightarrow{AC}$. Therefore, $(\vec{b}-\vec{a})\times(\vec{c}-\vec{a})$ is perpendicular to the plane.

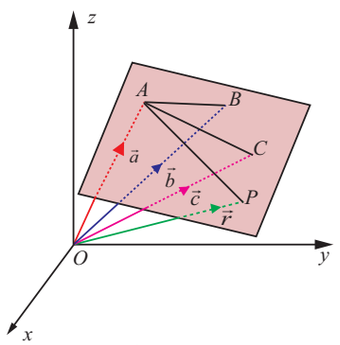



Fig. 6.28

If $\vec{r}$ is the position vector of an arbitrary point $P(x,y,z)$ on the plane, then the equation of the plane passing through the point $A$ with position vector $\vec{a}$ and perpendicular to the vector $(\vec{b}-\vec{a})\times(\vec{c}-\vec{a})$ is given by

$$(\vec{r}-\vec{a})\cdot((\vec{b}-\vec{a})\times(\vec{c}-\vec{a}))=\ 0\quad\text{or}\quad[\vec{r}-\vec{a},\ \vec{b}-\vec{a},\ \vec{c}-\vec{a}]=0$$

This is the non-parametric form of vector equation of the plane passing through three non-collinear points.

### (c) Cartesian form of equation

If $(x_1,y_1,z_1),(x_2,y_2,z_2)$ and $(x_3,y_3,z_3)$ are the coordinates of three non-collinear points $A,B,C$ with position vectors $\vec{a},\vec{b},\vec{c}$ respectively and $(x,y,z)$ is the coordinates of the point $P$ with position vector $\vec{r}$, then we have $\vec{a}=x_1\hat{i}+y_1\hat{j}+z_1\hat{k},\ \vec{b}=x_2\hat{i}+y_2\hat{j}+z_2\hat{k},\ \vec{c}=x_3\hat{i}+y_3\hat{j}+z_3\hat{k}$ and $\vec{r}=x\hat{i}+y\hat{j}+z\hat{k}$.



Chapter 6 Vector Algebra.indd 260 31-01-2020 17:52:30